

# Austin Senna Wijaya

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## Education

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| <b>BS</b> | <b>Columbia University</b> , Computer Science                                                                                                                                                                                                                                                                  | New York, NY        |
|           | <ul style="list-style-type: none"><li>• GPA: 4.22/4.0</li><li>• Full-tuition scholarship recipient.</li><li>• Dean's List.</li><li>• Relevant coursework: Natural Language Processing, Data Structures and Algorithms, Systems Programming, Linear Algebra and Probability, Competitive Programming.</li></ul> | Aug 2024 – May 2028 |

## Experience

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|  | <b>Columbia Data, Agents, and Processes Lab</b> , Data Agents Research Assistant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | New York, NY                     |
|  | <ul style="list-style-type: none"><li>• Optimized core evaluation infrastructure for question-answering agents over a 9.5 TB/~40M-document data lake, including parallel sandboxed benchmark workers, tool-call and reasoning telemetry, and BM25 and hybrid search engines.</li><li>• Improved GPT-5-mini semantic match from 2.22% to 56.3% by adding context compaction, loop-detection plugins, structured search-result context, and stronger data-analysis tools.</li><li>• Built an ablation framework that swaps in idealized search, planning, and data-analysis tools to isolate where agents fail on grounded question-answering over long context windows.</li><li>• Tech stack: Python, BM25, hybrid search, GPT-5-mini.</li></ul>        | Dec 2025 – present<br>10 months  |
|  | <b>Garner Health</b> , Software Engineer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | New York, NY                     |
|  | <ul style="list-style-type: none"><li>• Engineered a full-stack observability pipeline with automated dashboard and alert provisioning, using distributed tracing across Datadog RUM/Replay and OpenTelemetry (Loki, Prometheus, Tempo) to correlate frontend sessions with backend traces.</li><li>• Architected an autonomous on-call triage AI agent that ingests cross-stack telemetry to run root-cause analysis and generate verifiable fix pull requests under zero-PHI guardrails, resolving 10+ production incidents.</li><li>• Tech stack: Datadog (RUM, Replay), OpenTelemetry, Loki, Prometheus, Tempo.</li></ul>                                                                                                                          | June 2026 – Aug 2026<br>3 months |
|  | <b>Columbia Zuckerman Institute</b> , Machine Learning Research Assistant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | New York, NY                     |
|  | <ul style="list-style-type: none"><li>• Compressed Qwen2-Audio's 4096-D hidden states into 10-100D feature-specific representations, each preserving one linguistic or paralinguistic feature, such as emotion or reasoning, while suppressing the others.</li><li>• Curated training-ready datasets from 12 speech and language benchmarks, including converting text benchmarks into audio through parallel Kokoro-TTS generation.</li><li>• Built and parallelized a GPU-optimized PyTorch pipeline for Qwen2-Audio inference and hidden-state extraction, bottleneck-width sweeps, and supervised classification under cross-validation across 12 speech attributes and 120K+ audios.</li><li>• Tech stack: Python, PyTorch, Kokoro-TTS.</li></ul> | Dec 2025 – June 2026<br>7 months |

- Consolidated millions of payment events into user sessions in BigQuery, producing a product-level funnel used to diagnose major drop-off points.
- Automated large-scale Google Review scraping and sentiment analysis with Puppeteer and HuggingFace for continuous customer satisfaction tracking.
- Processed hundreds of thousands of tutor scheduling events with Apps Script and BigQuery to generate a real-time interactive Looker Studio tutor availability heatmap.
- Tech stack: Google Cloud (BigQuery, Looker Studio), Apps Script, Puppeteer, HuggingFace.

## Projects

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### **ResearcherX: Local-First AI Research IDE**

- Built a local-first AI research editor that answers against a bounded source library using graph-based RAG, flagging unsupported claims, contradictions, and citation drift in drafts.
- Implemented the retrieval and verification pipeline: turned PDFs, notes, and links into provenance-tracked source nodes, then checked draft paragraphs against relevant evidence with batched LLM calls.
- Engineered an asynchronous hybrid-search engine with Neo4j and LanceDB, using a dual-routing FastAPI and litellm pipeline to balance latency and cost between local and frontier models.
- Finalist (Top 6) at the Millard Chan Technology '99 Startup Competition 2026 and 2nd place at the Columbia Lion Cage Startup Competition 2026.
- Tech stack: Python (FastAPI), TypeScript (Next.js), Neo4j (Cypher), LanceDB, litellm, ProseMirror.

### **LakeAgent**

- Built infrastructure for deep-research agents to operate over both structured and unstructured data-lake sources.
- Designed an end-to-end pipeline for dataset discovery, automatic integration, and verifiable answer generation with explicit provenance.
- Applied the system to forecasting and analytic questions that combine structured signals with unstructured evidence.

### **Targeted Neural Audio Embeddings for Cortical Prediction**

- Built a brain-encoding pipeline that maps Qwen-Audio speech representations to cortical activations by extracting targeted task subspaces from large audio-language embeddings.
- Processed 150k+ audio examples across 15 benchmark datasets to isolate auditory signals such as emotion and reasoning.
- Performed dimensionality sweeps across model architectures to maximize task-relevant signal while suppressing noise.

### **Unstructured Cloud ELT Pipeline**

- Built an automated ELT pipeline to ingest chat screenshots into BigQuery using Vision AI for text extraction.
- Implemented sentiment and intent classification with BigQuery ML to support analytics and automated response workflows.
- Tech stack: BigQuery, Gemini API, Looker Studio, Vision AI, Python.

### **Untukmu Karyamu (Tencent Hackathon)**

- Placed 2nd at Tencent Kepler Plan S3 Competition 2025 with Best Code, Best Product Idea, and Popularity Award.
- Architected a generative web-builder with Next.js and Gemini API to automate deployment to Tencent EdgeOne for MSMEs.
- Tech stack: Next.js, Supabase, Puppeteer, EdgeOne Pages.

## Skills

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**Languages:** Python, SQL, JavaScript, Java, C, Rust

**Libraries and Frameworks:** Pandas, NumPy, PyTorch, Scikit-learn, Puppeteer, Selenium, BeautifulSoup, React, Next.js, Node.js

**Cloud and Databases:** Google Cloud (BigQuery, Looker Studio), AWS (EC2, S3, Bedrock), Supabase, PostgreSQL, MongoDB

**Tools:** GitHub, Docker, Claude Code, n8n, Copilot, Arduino, Notion

## Honors and Scholarships

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- **Databricks Grounded Reasoning Cup 2026:** Columbia representative; worked with Google.
- **Jane Street Mystery Planet Trading & Puzzle Competition 2026:** 1st Place.
- **Hack@Brown 2026:** Finalist and awarded 'Strongest Product Thinking,' working with Google Ventures and Partiful.
- **Millard Chan Technology '99 Startup Competition (2026):** Finalist (Top 6) with ResearcherX.
- **Columbia Lion Cage Startup Competition (2026):** 2nd Place with ResearcherX.
- **Tencent Kepler Plan S3 Competition 2025:** 2nd Place (Best Code, Best Product Idea, Popularity Award).
- **Clash of Champions Season 2:** Top 9 in an academic survival show for top Indonesian students.
- **Indonesia Maju Scholarship:** Full-ride scholarship recipient (Ministry of Education of Indonesia), 1 of 350 recipients.
- **55th International Chemistry Olympiad 2023:** Final stage (10th rank) for national team selection.